

Method of Moments

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So far, we have

- ① "guessed" the estimator for a parameter
- ② looked at what makes an estimator good.

But, what if we are interested in estimating some weird parameter

↳ physical significance is unknown.

Dist.	PDF	CDF	Mean & Variance
Normal/ Gaussian	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$	$F(x) = \Phi(x)$ Use z-table	$E[X] = \mu$ $\text{Var}[X] = \sigma^2$
Uniform ✓	$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$	$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{otherwise} \end{cases}$	$E[X] = \frac{1}{2}(a+b)$ ✓ $\text{Var}[X] = \frac{1}{12}(b-a)^2$
Exponential	$f(x) = \lambda \exp(-\lambda x)$	$F(x) = 1 - \exp(-\lambda x)$	$E[X] = \frac{1}{\lambda}$ $\text{Var}[X] = \frac{1}{\lambda^2}$
Gumbel ✓	$f(x) = \frac{1}{\beta} \exp[-z + \exp(-z)]$, where $z = \frac{x-\alpha}{\beta}$	$F(x) = \exp[-\exp(-z)]$ $\gamma = 0.577$ Euler-Mascheroni constant $\beta > 0$; scale parameter	$E[X] = \alpha + \beta\gamma$ $\text{Var}[X] = \frac{\pi^2}{6}\beta^2$ ✓ $\gamma = 0.577$
Lognormal	$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2\right]$	$F(x) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$ Use z-table	$E[X] = \exp\left(\mu + \frac{\sigma^2}{2}\right)$ $\text{Var}[X] = [\exp(\sigma^2) - 1] \exp(2\mu + \sigma^2)$

Method of moments asks you to

equal sample moments with the

corresponding population moment { which is what we need? intuitively doing }

• $E(x) = \frac{1}{n} \sum x_i$ OR $E(x) = \frac{1}{n} \sum x_i$

intuitively doing

OR

population

$$\left. \begin{aligned} E(X) &= \frac{1}{n} \sum_{i=1}^n x_i \\ E(X^2) &= \frac{1}{n} \sum_{i=1}^n x_i^2 \\ E(X^k) &= \frac{1}{n} \sum_{i=1}^n x_i^k \end{aligned} \right\} \text{sample}$$

$$E(X) = \frac{1}{n} \sum_{i=1}^n x_i$$

$$V(X) = E(X - E(X))^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$E(X - E(X))^k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k$$

Equate as many equations as many parameters starting from $E(X^1)$

e.g. 1 $X \sim \exp(\lambda)$

$$f_X(x; \lambda) = \lambda e^{-\lambda x} \quad ; x > 0$$

$$E(X) = \frac{1}{\lambda}$$

$$\bar{x} = E(X) = \frac{1}{\lambda}$$

$$\hat{\lambda}_{\text{mom}} = \bar{x}$$

$E[X^k] \leftarrow$
 $k^{\text{th}} \text{ sample moment} = \frac{1}{n} \sum_{i=1}^n X_i^k$

$E(\bar{x}) = \frac{1}{n} \sum_{i=1}^n x_i$

e.g. 2

Gumbel

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = E[X] = \alpha + \beta \gamma \quad - (1) \quad ; \gamma = 0.577$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = V(X) = \frac{\pi^2}{6} \beta^2 \quad - (2) \quad ; \beta > 0 \text{ scale param}$$

$$(2) \rightarrow \hat{\sigma}^2 = \frac{\pi^2}{6} \beta^2$$

$$\Rightarrow \hat{\beta}_{\text{mom}} = \frac{\sqrt{6} \hat{\sigma}}{\pi}$$

$$(1) \rightarrow \alpha + \beta \gamma = \bar{x} \Rightarrow \hat{\alpha}_{\text{mom}} = \bar{x} - \frac{\sqrt{6} \hat{\sigma}}{\pi} (0.577)$$

eg3

Uniform $U[a, b]$

$$\bar{x} = \frac{a+b}{2}, \quad \hat{\sigma}^2 = \left(\frac{b-a}{\sqrt{12}} \right)^2$$

$$\begin{aligned} a+b &= 2\bar{x} & \text{--- (1)} \\ -a+b &= 2\sqrt{3}\hat{\sigma} & \text{--- (2)} \end{aligned}$$

$$2b = 2\bar{x} + 2\sqrt{3}\hat{\sigma}$$

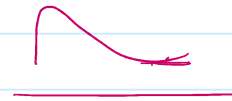
$$\begin{aligned} \hat{b}_{\text{mom}} &= \bar{x} + \sqrt{3}\hat{\sigma} \\ \hat{a}_{\text{mom}} &= \bar{x} - \sqrt{3}\hat{\sigma} \end{aligned}$$



Exponential distribution ; cont.

$$X \sim \exp(\lambda), \quad \lambda > 0$$

$$f_X(x) = \lambda e^{-\lambda x}, \quad \begin{matrix} x > 0 \\ \lambda > 0 \end{matrix}$$



$$E[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}$$

$V(X)$

$$U \sim [a, b]$$

$$f_X(x) = \frac{1}{b-a}, \quad b > a$$

Raw moments

k^{th}

$$= E[X^k]$$

Central moments

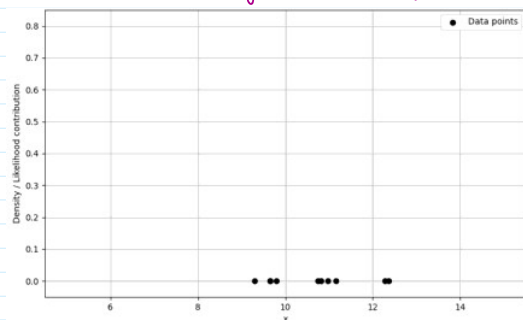
k^{th}

$$E[(X - E(X))^k]$$

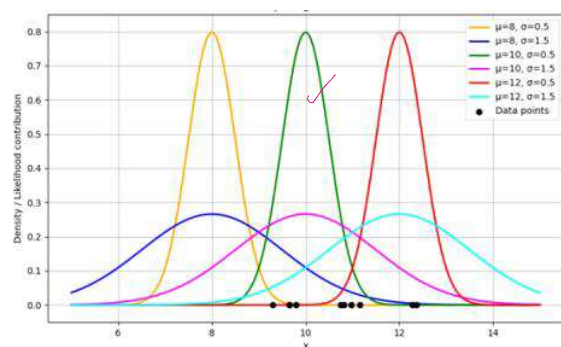
Mean = 1st raw moment

Variance = 2nd central moment

given the data, the objective is to find what values of distributional parameters were used to generate it?



Say, I know that these points came from normal density but not sure which μ & σ^2



Find $\mathcal{L}(\mu, \sigma | x_i) = f(x_i | \mu, \sigma)$ for each dart, x_i .
Maximize the product of likelihood,

$$\mathcal{L}(\mu, \sigma | \mathbf{x}) = \prod_{i=1}^n f(x_i | \mu, \sigma)$$

$$\hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} \mathcal{L}(\mu, \sigma | \mathbf{x})$$

$$\hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} \ln[\mathcal{L}(\mu, \sigma | \mathbf{x})]$$

More generally

given the data, find the values of the parameters that maximize the likelihood function.

e.g. Consider $x_1, \dots, x_n \sim \exp(\lambda)$

$$f_X(x; \lambda) = \lambda e^{-\lambda x} \quad ; \quad x > 0$$

$$L(\lambda) = \prod_{i=1}^n f(x_i) = (\lambda e^{-\lambda x_1}) (\lambda e^{-\lambda x_2}) \dots (\lambda e^{-\lambda x_n})$$

$$\ln L(\lambda) = n \ln \lambda - \lambda \sum x_i$$

$$\frac{\partial \ln L(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum x_i = 0 \Rightarrow \hat{\lambda}_{MLE} = \frac{1}{\bar{x}}$$

$$\frac{\partial^2 \ln l(\lambda)}{\partial \lambda^2} = -\frac{n}{\lambda^2} < 0$$

$$\hat{\lambda}_{MLE} = \hat{\lambda}_{MOM}$$

Ex. Do it for Normal(μ, σ^2) for MOM & MLE

ML estimators have many nice properties

① Bias $\rightarrow 0$ as $n \rightarrow \infty$ [Asymptotically unbiased]

② Variance property is also satisfied in limit

③ consistent

④ Asymptotic normality under some mild regularity conditions

Invariance Property of MLE

for e.g. $\bar{X}$ is MLE of θ for normal distⁿ of known

What is MLE of $\theta + 20$?

$$= \bar{X} + 20$$

$$\Rightarrow f(\theta) \rightarrow f(\hat{\theta}_{MLE})$$

$X \sim U[0, \theta]$. What is the MLE of θ ?

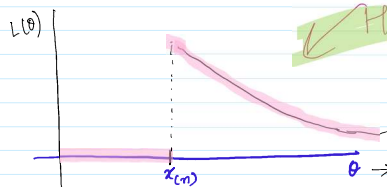
$$f_X(x; \theta) = \frac{1}{\theta} \quad ; \quad 0 \leq x \leq \theta \quad \checkmark$$

$$L(\theta) = \frac{1}{\theta} \cdot \dots \cdot \frac{1}{\theta} = \frac{1}{\theta^n} I_{(0 < x_1 < \theta)} \cdot I_{(0 < x_n < \theta)}$$

$$L(\theta) = \frac{1}{\theta^n} I_{[0 < x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)} \leq \theta]}$$

$$\text{ANSWER } \hat{\theta}_{MLE} = X_{(n)} = \max\{X_1, \dots, X_n\}$$

DERIVE



HINT